

Grace — LANE A: BLIND verification of “up sits on even rungs” (verify from geometry, not from what breaks CKM)

2026-08-09. Casey’s discipline: verify up-even independent of the mixing. Result: the parity SPLIT is forced; but “up on even rungs” is a MIXING-parity statement, not a mass-position claim — and conflating them fails a blind check.

What IS geometric and forced (the split = the breaking)

The Q^5 cohomology ring $\mathbb{Z}[h]/h^6$ has a $\mathbb{Z}_2$ grading (even/odd degree = the boundary-fold parity). Up-type = EVEN classes $\{h^0, h^2, h^4\}$, down-type = ODD $\{h^1, h^3, h^5\}$ (T2544). This parity split is forced by the geometry (the ring’s grading), independent of CKM. Its consequence: the naive $\langle \text{up} | \text{down} \rangle$ overlap vanishes (even ∇ odd) $\rightarrow$ mixing MUST be W-mediated (J_W = the degree- ± 1 cup-with-h operator, the unique even $\leftrightarrow$ odd bridge = the SM charged current) $\rightarrow$ small near-diagonal CKM. The breaking is geometric — Structure-Derived (T2544). This is the honest content of “the mixing is the up/down breaking.”

★ THE CATCH (blind check — do NOT conflate parity with mass positions)

If “up sits on even rungs” meant up MASSES sit at even degrees $\{0, 2, 4\}$, the ladder would be $(N_c)_\lambda$ at $\lambda=0, 2, 4$ = 1:12:360. Observed u:c:t $\approx$ 1:588:79981. WRONG — the even-degree FK ladder is not the up mass spectrum. Up mass is top-saturation / non-FK (F822, K1012 — up is radial/boundary, not a degree-ladder).

Contrast the down: odd degrees $\{1, 3, 5\} \rightarrow (N_c)_\lambda = \{3, 60, 2520\}$ — matches the down ladder AND gives $V_{us}=1/\sqrt{20}$. So: - Down = odd works for BOTH parity AND mass ($\{1, 3, 5\}$). - Up = even works for the cohomology PARITY (mixing) but NOT the mass (up is saturation).

So “up on even rungs” is a mixing-parity statement (cohomology, T2544), not a mass-position claim. The up/down asymmetry is actually TWO forced facts: (a) opposite cohomology parity (the mixing engine), (b) down=degree-ladder vs up=saturation (the mass). Both forced; different mechanisms. Don’t test the parity-overlap by putting up on even MASS degrees — that gives the wrong up masses.

Is up=EVEN (vs up=odd) blind-forced, or assigned?

- The split (up ∇ down) is forced (the $\mathbb{Z}_2$ grading). ∇
- The specific label (up is the EVEN one) rides a candidate anchor: the Higgs VEV is the degree-0 (even, h^0) SO(5)-singlet zero-mode (F338), so the sector aligned with the even VEV is even. That is a plausible geometric anchor but I did NOT find a clean blind forcing that UP (not down) is the even one beyond the VEV-parity reading. up=even-specifically = candidate-anchored (VEV parity), not fully blind-forced. (Physically $|CKM|$ is invariant under swapping the even/odd labels, so the mixing test doesn’t depend on which is even — only that they’re opposite.)

★ Blind score for the parity-overlap $\rightarrow$ CKM test (armed)

When @Lyra/@Elie test whether the even/odd cross-parity overlap gives observed CKM with no fitted coefficient: 1. Use the COHOMOLOGY parity (T2544: even $\{h^0, h^2, h^4\}$ vs odd $\{h^1, h^3, h^5\}$), NOT up-on-even-MASS-degrees (that fails the mass check above). 2. PASS = the cross-parity J_W overlap gives the CKM pattern with NO fitted coefficient (the CG/parity structure alone) $\rightarrow$ ties the mixing to the oddness engine (same odd N_c , odd n_C as $\sin^2\theta_W$ and CP). 3. Keep the split-is-forced (Derived) separate from the exact-values (Candidate): $V_{us}=1/\sqrt{20}$ stands Derived (down odd ladder); the CKM PATTERN rides the up/down asymmetry — Candidate until the overlap gives it coefficient-free. A hand-picked coefficient banks nothing.

Status

No node (a blind verification + a conflation catch). Net: the parity SPLIT (the breaking) is geometrically forced and IS the mixing engine; up=even holds for the cohomology parity (mixing) but NOT the mass positions

(up=saturation); up=even-specifically rides the degree-0 VEV parity (candidate anchor). The oddness engine (odd N_c , odd n_C) that runs through $\sin^2\theta_W$ and CP plausibly runs through the mixing too — the coefficient-free overlap test decides it. Cross-refs: T2544 (parity shape), F822/K1012 (up=saturation, down={1,3,5}), F338 (VEV=degree-0 even), `grace_flavor_is_linear_algebra`. Nothing pushed.